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Key Points:

- We infer anisotropic ice fabric inside the ice stream of Greenland's fastest flowing outlet glacier using seismic shear wave splitting
- The measured horizontal cone fabric indicates that deformation within the ice stream is significantly easier along-flow than across-flow
- This observed fabric highlights that assuming isotropic ice within ice flow models could lead to significant inaccuracies

Supporting Information:

Supporting Information may be found in the online version of this article.

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Interpreting Ice Anisotropy of Greenland's Fastest Outlet Glacier Using Basal Icequakes

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Abstract Ice anisotropy can have significant impact on ice rheology and flow dynamics. However, a lack of in situ measurements limits the validation and incorporation of anisotropy, so most ice flow models assume isotropic ice. Especially observations of anisotropy in fast-flowing ice streams are scarce. Here, we present an assessment of anisotropic ice fabric inside a fast-flowing Greenlandic ice stream. Seismic anisotropy, measured through shear wave splitting in basal icequakes, reveals an anisotropic horizontal cone fabric. The corresponding fabric tensor suggests that ice likely deforms at least seven times easier along-flow than across-flow. The presence of such a strong anisotropic fabric within the fast-flowing ice stream emphasizes that assuming isotropic fabric in ice flow models likely introduces significant inaccuracies. Furthermore, variations between our observations and measurements from other regions highlight the necessity of comprehensive in situ data to better understand spatial variations in ice anisotropy and how fabrics develop.

Plain Language Summary Glacier ice consists of countless small hexagonal crystals. Initially, as snow compacts under its own weight, these crystals are randomly oriented, forming isotropic ice, which deforms equally in all directions. However, as the glacier flows, compression and stretching causes the crystals to align in specific patterns, creating anisotropic ice, which deforms more easily in certain directions. These specific patterns are referred to as ice fabric and influence the glacier flow. Despite its impact on ice flow most models still assume that ice is isotropic, as a lack of direct measurements of ice anisotropy limits the possibility of including it. In this study, we infer the ice anisotropy inside the ice stream of Greenland's fastest flowing outlet glacier by using seismic measurements of small icequakes occurring at the glacier bed. The results indicate that the ice inside the ice stream deforms much more easily along the flow direction than across it. This highlights the need to include ice anisotropy into models to improve accuracy and underscores the importance of more measurements of ice anisotropy in fast-flowing regions.

1. Introduction

Glacier ice forms as snow is compressed under its own weight, initially forming isotropic ice, meaning the ice crystals are randomly oriented. Over time, under sustained overburden stress and continued creeping deformation, ice crystals will start to align, creating anisotropic ice (Alley, 1988; Faria et al., 2014; Qi et al., 2019). The re-alignment of ice crystals occurs due to the intrinsic anisotropy of hexagonal ice crystals, which preferentially deform along the basal plane of the crystal (Duval et al., 1983). This preferred deformation plane causes the crystals to rotate under stress, developing crystal orientation fabric (COF), also called ice fabric. The type of ice fabric influences the mechanical properties of the ice and consequently its directional sensitivity to stress (Alley, 1988, 1992), ultimately affecting ice flow patterns. There is general agreement that ice anisotropy likely plays a significant role in ice flow dynamics and that neglecting it introduces inaccuracies into ice flow models (e.g., Azuma, 1994; Graham et al., 2018; Ranganathan & Minchew, 2024; Rathmann & Lilien, 2022). However, the most commonly used flow law (Glen's flow law) describes isotropic ice and is not valid for anisotropic ice (Glen, 1958; Nye, 1952). While efforts have been made to integrate ice anisotropy into flow laws and models, progress is restricted by the lack of in situ measurements of ice fabric. This limitation is particularly present in dynamic, fast-flowing regions such as ice streams and outlet glaciers, which are the primary channels for solid mass loss (Gerber et al., 2023; Graham et al., 2018).

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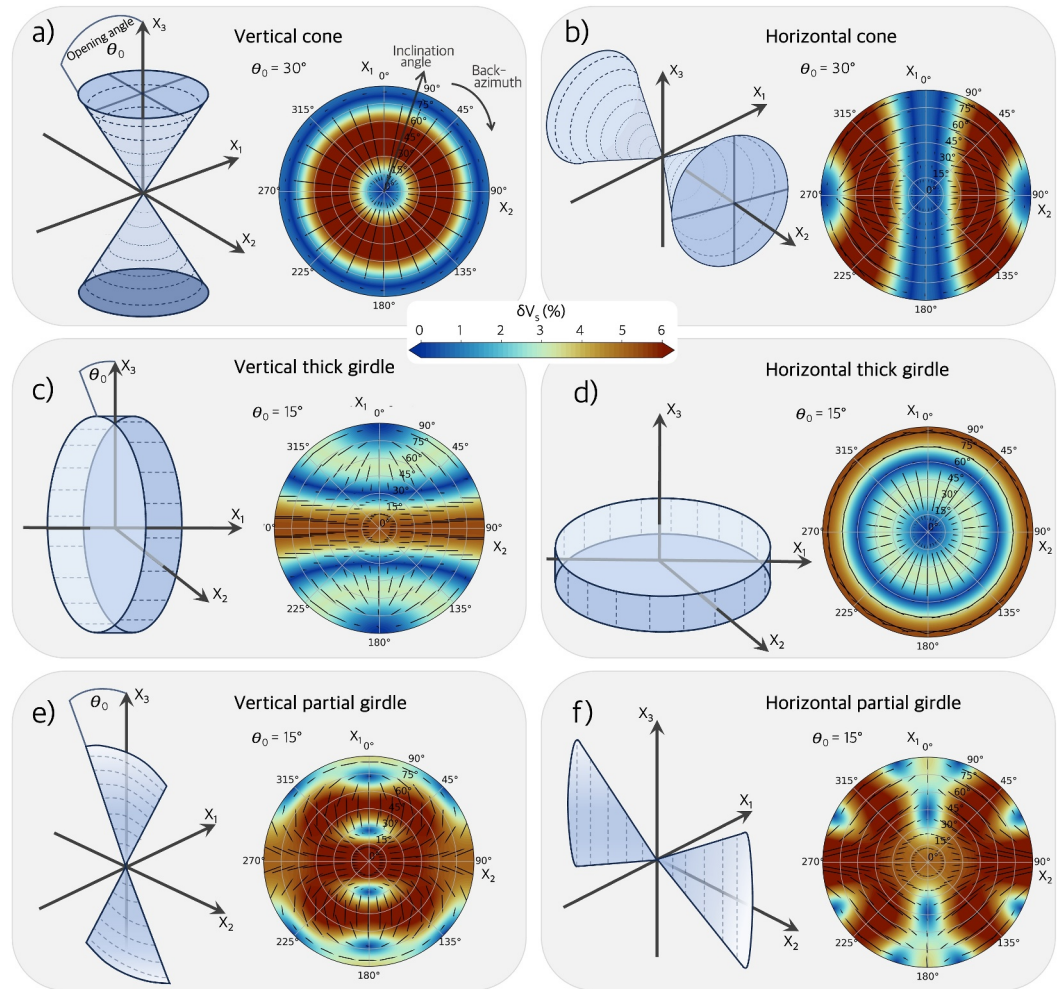


Figure 1. Overview of ice fabrics with statistical c -axis representations, and modeled shear-wave anisotropy patterns (δV_s) on polar diagrams (adapted from Kufner et al., 2023; Maurel et al., 2015; Smith et al., 2017). Blue geometric diagrams illustrate the envelope of c -axis orientations, with the opening angle θ_0 shown for the vertical fabric, corresponding equivalently to the horizontal version. Polar diagrams depict shear-wave anisotropy strength, δV_s (background color) for ray paths with back-azimuth (angular axis) and incidence angle (radial axis). Black lines in the polar diagrams represent the fast s -wave orientation (line orientation) and anisotropy strength (line length). (a) Vertical and (b) horizontal cone fabric ($\theta_0 = 30^\circ$). (c) Vertical and (d) horizontal thick girdle fabric ($\theta_0 = 15^\circ$). (e) Vertical and (f) horizontal partial girdle fabric ($\theta_0 = 15^\circ$).

The different ice fabrics can be described by a statistical representation of the orientation of the c -axes, which is the axis perpendicular to the hexagonal ice crystal rings (Duval, 1981). Typically, the c -axes tend to rotate away from the direction of extension and toward the direction of compression. Different stress regimes give rise to distinct ice fabrics. The common fabrics observed in glaciers, which are later used for model inversion, are shown in Figure 1. Most in situ observations of ice fabric are made on ice cores, commonly drilled at ice domes or divides, where the ice flows slowly and vertical compression and horizontal simple shear are dominant, typically producing a vertical cone fabric (Figure 1a) (Faria et al., 2014; Weikusat et al., 2017). However, this setting is not representative of stress conditions in ice streams and outlet glaciers, where strong lateral and along-flow stresses are present. In situ measurements in these regions are challenging and therefore scarce.

Each ice fabric creates a distinct pattern of anisotropy in elastic and electromagnetic wave-speeds (Diez & Eisen, 2015; Diez et al., 2015; Maurel et al., 2015). Geophysical methods, such as seismology and radar sounding, can utilize these patterns to infer ice fabric, offering an alternative for dynamic regions where ice core drilling is prohibitive. However, most previous studies focus on Antarctic ice streams (e.g., Harland et al., 2013; Jordan et al., 2019; Kufner et al., 2023; Smith et al., 2017) while ice fabric measurements from Greenland ice streams

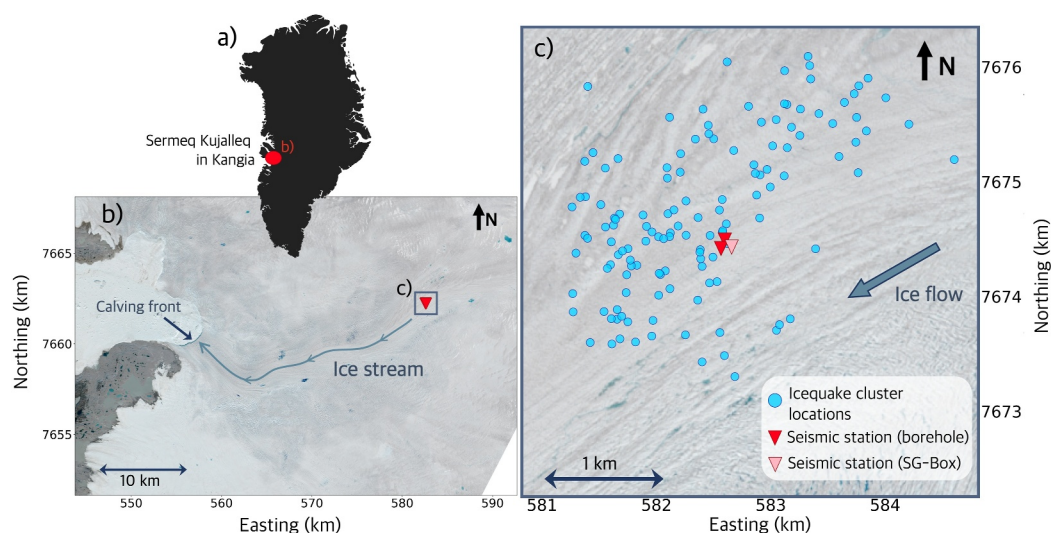


Figure 2. Overview of the study site and seismic station locations. (a) Map of Greenland indicating Sermeq Kujalleq in Kangia (SKK) with red dot. (b) Overview of the fast ice stream, with the calving front on the west. The seismic array (red triangles) and ice stream including flow direction (blue arrow line) are indicated. (c) Zoomed-in view of seismic array (red triangles) and basal icequake cluster locations (blue dots). The array includes two borehole sensors (West and Northwest) and one self-sufficient seismic box (Southeast) (Nap et al., 2023). Arrow indicates ice flow direction at seismic array. Background image from Sentinel-2 (ESA, 2023).

remain limited. The few existing studies on Greenlandic ice streams do not directly resolve ice fabric and are typically conducted far upstream (Gerber et al., 2023) or outside the main ice stream (Booth et al., 2020; Horgan et al., 2008).

Here, we present an assessment of ice fabric inside Greenland's fastest-flowing outlet glacier. By using seismic shear wave anisotropy observed in basal icequakes recorded within the main ice stream, we provide, to our knowledge, a first full estimate of ice fabric inside such a fast-flowing Greenlandic glacier. These observations allow us to interpret the ice anisotropy in these flow conditions and its effect on ice flow dynamics.

2. Study Site and Basal Icequake Data Set

The data set used in this study was derived from a three-station seismic array deployed in the center of the fast-flowing trunk of Sermeq Kujalleq in Kangia (SKK; also known as Jakobshavn Isbræ), an outlet glacier located on Greenland's west coast (Figures 2a and 2b). The array was part of a larger field experiment conducted in 2022 (Wehrle et al., 2025). This outlet glacier is the fastest flowing outlet glacier in the world, reaching maximum velocities of 40 m d^{-1} in summer (Joughin et al., 2008, 2020). Bed topography maps show that the roughly east-west oriented fast-flowing trunk overlies a deep fjord where the ice reaches a maximum thickness of approximately 2,000–2,500 m, tapering to approximately 1,000 m thickness toward the margins (An et al., 2017; Clarke & Echelmeyer, 1996). Largely due to this topography, the fast flowing trunk is neighbored by much slower flowing ice on the north and south side creating a laterally confined stream. The high stresses created by this velocity difference give rise to heavily crevassed shear margins. With exception of the final 6 km toward the calving front, we observe that the center fast-flowing trunk exhibits relatively sparse crevassing.

The seismic array consisted of two borehole sensors (4.5 Hz Geospace ONE) and one self-sufficient seismic box, an autonomous seismic sensor specifically developed for these deployments (SG-box, 14 Hz SM-6 Omni-Directional, see Nap et al., 2023), set-up in a triangle shape with a sensor spacing of 80–100 m (Figure 2c). The borehole sensors were equipped with a Nanometrics Pegasus digitizer sampling at 500 Hz and the SG-box with a DIGOS dataCUBE³ sampling at 400 Hz. The borehole sensors were drilled approximately 3 m into the ice. The SG-boxes were placed directly on top of the ice. No firn was present at the time of deployment.

Between 21 May and 11 July 2022, approximately 25,000 basal icequakes were recorded at this array. Singular events were found through initial visual inspection and expanded to clusters through cross-correlation template

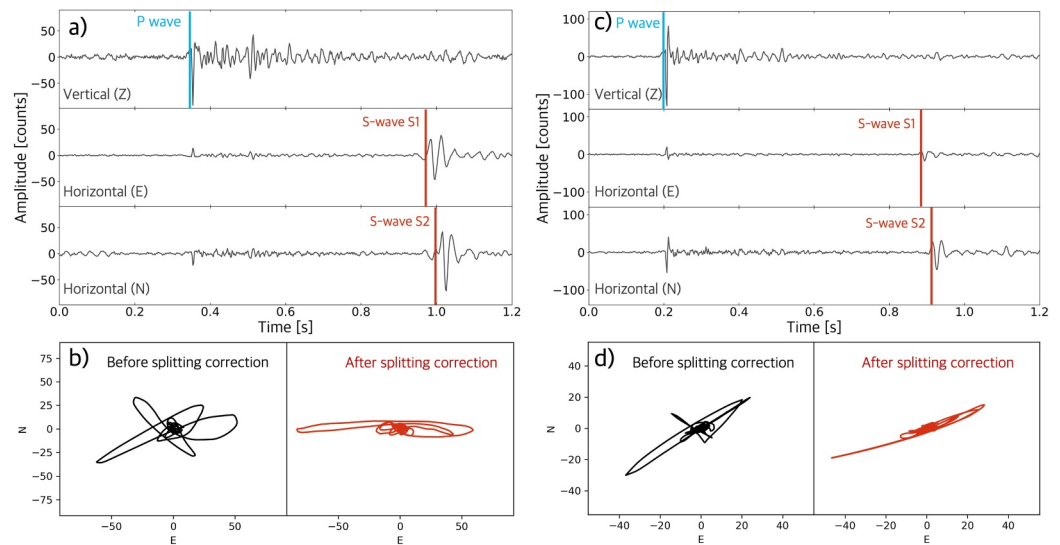


Figure 3. Examples of average cluster waveforms and present shear-wave splitting. (a) and (c) Average waveforms of two clusters showing clear shear-wave splitting (S1 arriving earlier than S2). (b) and (d) Particle motions in map view (E-N component), of the split S-waves (in black) and after splitting correction (in red).

searches, as basal (stick-slip) icequakes typically cluster spatially and emit nearly identical waveforms (e.g., Gräff et al., 2021; Roeoesli et al., 2016; Smith et al., 2015). In total, we identified 133 clusters containing between 3 and 2,500 events, which exhibited varying degrees of shear wave splitting, a clear indication that the waves traveled through an anisotropic medium (see Section 3.1). The cluster locations, determined using NonLinLoc (Lomax et al., 2000), are displayed in Figure 2c. All clusters locate around the ice-bed interface from An et al. (2017), more details on the locations and uncertainties are presented in Supporting Information S1. Given the nearly identical waveforms within clusters and the spatial clustering of the icequakes, we treated the average stacked waveform from each cluster as a single measurement, resulting in a maximum total of 399 shear wave splitting measurements (i.e., 133 event-station pairs per sensor).

3. Anisotropy Measurements and Ice Fabric Inversion

3.1. Shear Wave Splitting Analysis

A characteristic of basal stick-slip icequakes is that they emit strong P- and S-waves (Deichmann et al., 2000). When traveling through anisotropic ice, S-waves split into two orthogonal components, each traveling at different speeds, a concept known as shear wave splitting (SWS) or seismic birefringence (Crampin, 1985). Examples of split S-waves are shown in Figures 3a and 3c. The delay time between the fast and slow S-waves indicates the strength of anisotropy along the ray path, while the fast S-wave polarization reflects the fabric's symmetry. By determining the best-fitting delay time (δt) and fast S-wave polarization direction (ϕ) for each event-station pair that removes the SWS effect, elastic anisotropy can be mapped across different azimuths and inclination angles. The resulting shear wave anisotropy pattern provides insights into the most likely ice fabric, as illustrated by the shear wave anisotropy patterns in Figure 1.

To determine the best-fitting δt and ϕ for each event-station pair, we used the SWSPy software package, which enables fully automated analysis. It includes quality metrics that support the selection of high-confidence measurements. Specifically, it minimizes the ratio of the second to first eigenvalue (λ_2/λ_1), which quantifies the linearity of the split S-wave's particle motion and serves as a value for measurement quality (Hudson et al., 2023). Examples of split and corrected S-wave particle motion are shown in Figures 3b and 3d. Before analysis, the waveforms were rotated into a geographical reference frame (East, North, Vertical) from field orientation using the icequake locations. The search ranges are 0–0.1 s for δt and 0–180° for ϕ .

Using this method, 399 event-station pairs were analyzed based on average cluster waveforms (Section 2). To avoid bias from manual selection, we used the eigenvalue ratio and δt error provided by SWSPy to identify high-

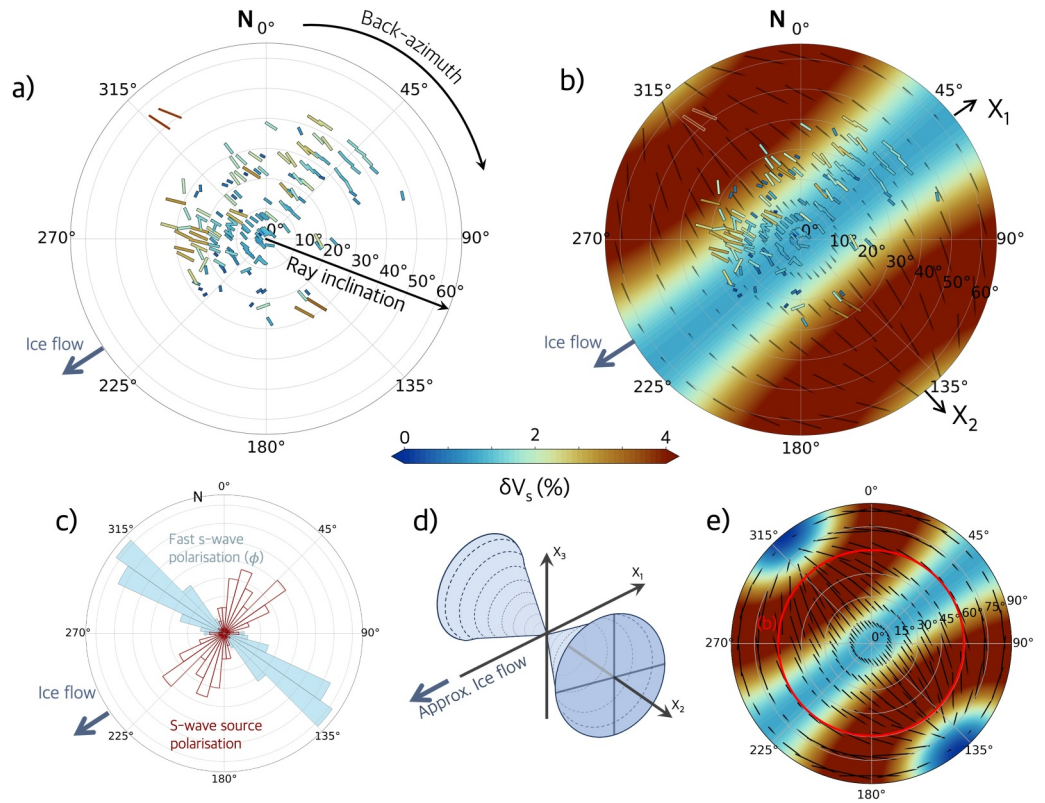


Figure 4. Shear wave splitting results and best-fitting fabric model. (a) SWS measurements (bars) plotted at the station-to-event back-azimuth (from North) and ray inclination, where 0° is vertical and 90° is horizontal. The inclination axis is limited to 60° for visibility. Measurements plotted in 5° × 5° azimuth and inclination bins. Bar orientation indicates ϕ (fast s-wave polarization), length and color represent δV_s (percentage of anisotropy along the ray-path). Ice flow is shown by the blue arrow. (b) Best-fitting fabric model: horizontal cone fabric with an opening angle $\theta_0 = 38^\circ$, rotated 51° around X_3 (see (d)). Splitting measurements from (a) are overlaid for comparison. (c) Rose diagram of ϕ (blue) and initial s-wave polarization before splitting correction (red), both in 10° bins. (d) Statistical representation of c -axes orientations of horizontal cone fabric. (e) Fabric model from (c) with the inclination axis extended to 90°.

confidence measurements. Thresholds of 0.15 for the eigenvalue ratio and 0.007 s for δt error were chosen based on visual inspection of a representative subset of ~40 measurements. Additionally, five measurements showing clear cycle skipping were manually removed. This resulted in a total of 254 retained SWS measurements. A larger number of the selected measurements come from the two borehole stations as the self-sufficient seismic box is more susceptible to noise, especially in strong wind conditions, resulting in less usable SWS measurements (Nap et al., 2023). For each SWS measurement the strength of S-wave anisotropy along the ray-path can be expressed as a percentage using:

$$\delta V_s = 100 \times \left(\frac{V_s \cdot \delta t}{r} \right) \quad (1)$$

where V_s is the average S-wave velocity of 1,950 ms⁻¹ (Walter et al., 2014), δt the best-fitting delay time, and r the event-station distance. These measurements, along with the fast S-wave direction, can be plotted on a polar plot at the corresponding azimuth and ray incidence of that ray-path. In Figure 4a, the results are plotted as averages in vertical incidence and back-azimuth bins of 5° × 5°, a polar plot of all singular measurements can be found in Figure S2 in Supporting Information S1. The pattern we observe is that along the flow line (approximately NE-SW) generally lower anisotropy (0.5%–1.5%) is measured and, although limited coverage, we measure higher anisotropy outside of this axis (above 2.5%). The dominant polarization direction of the fast S-wave is nearly perpendicular to the flow direction at an average of ~311° (Figure 4c in blue). The initial S-

wave source polarization is approximately along flow at $\sim 50^\circ$ average (Figure 4c in red), which matches the likely source mechanism of the ice slipping over the bed (Kufner et al., 2021; Roeoesli et al., 2016).

3.2. Inversion for Ice Fabric

To determine the ice fabric that best explains the observed shear wave splitting pattern, we performed an inversion using modeled S-wave anisotropy and polarization values for the fabric types shown in Figure 1. These models are based on the elasticity tensors corresponding to each fabric from Maurel et al. (2015). By applying the Christoffel equation to the fabric-specific elasticity tensor, we computed phase velocities and fast S-wave polarization angles across all azimuths and inclination angles (Wookey, 2012). The resulting phase velocities for the fast and slow S-wave are then used to calculate the S-wave anisotropy percentage (δV_{sM}).

The inversion was performed by summing the misfits between each measured and modeled data point. The global misfit is defined by:

$$f = \sum_{i=1}^n \sqrt{\left(\frac{\Phi_i - \Phi_{Mi}}{\max(\Phi) - \min(\Phi)} \right)^2 + \left(\frac{\delta V_{si} - \delta V_{sMi}}{\max(\delta V_s) - \min(\delta V_s)} \right)^2} \quad (2)$$

where n is the total number of SWS measurements, Φ_i and Φ_{Mi} are the observed and modeled orientations of the fast S-wave and δV_{si} and δV_{sMi} are the observed and modeled values for the anisotropy, respectively. To avoid bias toward Φ or δV_s , the differences between observed and modeled data are normalized by the total range of Φ and δV_s , respectively.

All ice fabrics from Figure 1 were included in the inversion and the misfit was calculated for each possible opening angle ($0-90^\circ$) and rotation around the X_3 axis ($0-180^\circ$, as the fabrics have a symmetry axis along X_1 and X_2). We also tested the fabrics with rotations around the X_1 and X_2 axes, however, this did not reduce the misfit further and yields highly asymmetric anisotropy patterns that cannot reliably be resolved with the data coverage. We chose not to test more complex “mixed” fabrics, as done in Smith et al. (2017) and Kufner et al. (2023), to prevent overfitting the data with limited measurements. To prevent bias toward regions with a higher measurement density, the data were grouped into $5^\circ \times 5^\circ$ bins based on incidence and azimuth angles.

Among the tested fabrics, the best fit to the observed SWS measurements corresponds to a horizontal cone with an opening angle of $\theta_0 = 38^\circ$, rotated by 50° around the X_3 axis (Figures 4b, 4d, and 4e). The global misfit of this fabric is 41% lower than that for a fully isotropic fabric (30 compared to 74) and 37% lower than the lowest misfit obtained for any of the other tested fabrics. This fabric is generally created by simple shear along the X_1 - X_3 plane or uniaxial compression along the X_2 axis (Alley, 1988; Diez & Eisen, 2015). More information on the misfits and local minima can be found in Figure S3 in Supporting Information S1.

4. Discussion

The horizontal cone fabric at Sermeq Kujalleq (SKK) likely results from strong vertical shear induced by the fast-flowing, laterally confined ice stream. Ice crystals tend to rotate such that their basal planes—where deformation preferentially occurs—align with the shear plane, which in this case follows the ice flow, leading to c -axes roughly perpendicular to flow. At the seismic array (Figure 2), the main trunk of the ice stream is ≈ 5 km wide, bordered by heavily crevassed shear margins. This geometry is probably key in the formation of this specific anisotropic fabric. The sparse crevassing in the main trunk suggests that strain from the strong velocity gradient between the fast and slow ice is accommodated mainly through internal deformation and/or basal slip (Lüthi et al., 2002). Furthermore, flow line tracking shows that lateral shearing is stronger than along-flow strain to some degree for at least 30–40 km upstream (Figure S4 in Supporting Information S1), indicating that the observed anisotropy developed progressively over an extended distance.

Despite the limited data coverage, the fit between the observed data and the horizontal cone fabric is strong. This can be explained by the fact that the horizontal cone fabric with opening angles between 24° – 43° is the only fabric that reproduces the observed pattern of low anisotropy along the flow line and higher anisotropy elsewhere. Additionally, the fast S-wave polarization of the modeled fabric closely aligns with observations, predominantly perpendicular to ice flow (Figure 4). It is important to note, however, that the inferred ice fabric represents a

depth-averaged fabric over the full ice thickness. In reality, depth variations may include multiple fabric layers, with the horizontal cone dominating the signal (Booth et al., 2020; Fichtner et al., 2023; Horgan et al., 2008; Kufner et al., 2023). Given the array size, the number of events and resulting SWS data coverage, it is difficult to resolve more complex layering of fabrics. The uncertainty in the locations of the icequake clusters (see Figure S1 in Supporting Information S1) introduces some degree of uncertainty in the placement of observations on the polar plot. However, this does not significantly affect the strength of observed anisotropy or the fast S-wave orientation.

While this exact fabric has not been observed in fast-flowing outlet glaciers, similar directional anisotropy has been documented in the shear margins of the North East Greenland Ice Stream (NEGIS) (Gerber et al., 2023) and the floating shear margins of Priestly glacier, Antarctica (Thomas et al., 2021). Modeling studies also suggest comparable fabrics under similar deformation regimes (Lilien et al., 2021). SWS studies on Rutford Ice Stream (Kufner et al., 2023; Smith et al., 2017), do not show the same pattern. They find a combination or layering of fabrics—including a vertical cone and either a horizontal partial girdle or vertical thick girdle. This discrepancy is likely due to the contrasting glaciological settings: SKK experiences significantly faster flow and greater lateral confinement than the Rutford Ice Stream sites, which likely drives the development of different anisotropic fabrics. The only ice fabric study near SKK (Horgan et al., 2008) hypothesizes mainly vertical *c*-axis orientations based on seismic reflection data. Since their study site is outside the main ice stream, a different fabric from that within the fast-flowing trunk is to be expected. These observational differences underscore regional variability in ice fabric, and suggest that repeating this study at a different location within or outside the ice stream would likely reveal different patterns.

A simple way to assess the impact of the ice fabric on ice flow dynamics is through the fabric tensor (Woodcock, 1977). Using equations from Maurel et al. (2015), we can compute its eigenvalues, which represent the principal *c*-axis orientations. This tensor is commonly used in ice sheet modeling to derive viscosity matrices and enhancement factors (Gerber et al., 2023; Gillet-Chaulet et al., 2006; Jordan et al., 2022; Ma et al., 2010; Montagnat et al., 2014). In a fully isotropic fabric, all three eigenvalues are equal to 1/3, indicating no preferred crystal orientation. For a horizontal cone fabric with $\theta_0 = 38^\circ$, the eigenvalues are $\lambda_1 = 0.11$, $\lambda_2 = 0.78$, and $\lambda_3 = 0.11$, indicating a preferred *c*-axis direction along X_2 . Given the dependence of the viscosity tensor on the fabric tensor (Gerber et al., 2023; Gillet-Chaulet et al., 2005, 2006; Jordan et al., 2022), we can estimate from the fabric tensor ratio $\lambda_2/\lambda_{1,3} = 7.1$, that deformation along-flow is likely at least seven times easier than across-flow. This could significantly influence the glacier's response to stress perturbations.

Although ice anisotropy is increasingly included in ice flow models (Gillet-Chaulet et al., 2005; Graham et al., 2018; Lilien et al., 2021), direct constraints on fabric for validation and integration remain limited. Incorporating in situ fabric measurements from fast-flowing regions, like those presented here, could refine model parameterization and reduce errors in predicting ice sheet evolution. Further advancements, such as borehole seismology in dynamic regions, combined with radar and/or SWS measurements, would improve understanding of anisotropy variation with depth and fabric layering (Booth et al., 2020; Brisbourne et al., 2021; Fichtner et al., 2023; Kufner et al., 2023).

While most outlet glaciers and ice streams flow slower than SKK, similar fabric is likely representative of laterally confined ice streams with a narrow fast-flowing trunk, bordered by slower ice. Such ice streams are common in Greenland and Antarctica, where they essentially flow through subglacial fjords shaped by erosion. As these ice streams are the main channels through which the ice sheets transport ice mass toward the ocean, understanding how ice fabric influences their flow dynamics is essential for accurate prediction of ice sheet evolution.

5. Conclusions

This study presents a first full estimate of anisotropic ice fabric within a fast-flowing laterally confined outlet glacier in Greenland and is likely representative of similar ice streams. Through shear wave splitting analysis of basal microseismic icequakes, we infer a horizontal cone fabric with a 38° opening angle, indicating along-flow deformation is significantly simpler than across-flow. This fabric differs from the more vertical alignments observed in ice cores (e.g., Faria et al., 2014) and hypothesized in studies on the slower-moving ice (Horgan et al., 2008). It also contrasts to a certain degree with observations using similar methods at Rutford Ice Stream,

Antarctica (Kufner et al., 2023; Smith et al., 2017), while our observations do resemble fabrics found at NEGIS, Greenland (Gerber et al., 2023), and in the floating shear margin of Priestly Glacier, Antarctica (Thomas et al., 2021). The differences in these observations clearly highlight the regional variability of ice fabric within and across ice sheets and underscore the inaccuracies that are likely introduced in ice flow laws and models by assuming isotropic ice.

Given that ice streams and outlet glaciers are the main pathways for solid mass loss, more in situ measurements in these dynamic regions are essential. Such observations are necessary for validating models, integrating ice fabric into large-scale ice flow simulations, and ultimately improving predictions of ice sheet evolution.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

The seismic data used in this study are publicly available through the data portal of the Swiss Seismological Service at ETH Zürich (<http://networks.seismo.ethz.ch/en/networks/ch/>) under the doi from Lüthi and Walter (2021). The network name is 3Z and the stations analysed in this study are SK11G, SK12G, and SK12S, where SK12S is the SG-Box and the others are borehole geophones. More information on how to download the data from the network can be found at <http://www.seismo.ethz.ch/en/home/> under the menu *Knowledge* → *Earthquake data & web portals*. The software package SWSPy used for shear wave splitting analysis is publicly available at <https://swspy.readthedocs.io/en/latest/> and described in Hudson et al. (2023). A Python package for modeling seismic anisotropy and performing the inversion, IceAnisotropy, is currently under development and planned for public release. The methodology is detailed in Section 3.2 and relies on elasticity tensors as described in Maurel et al. (2015). While the package itself is not yet publicly available, all methods are described using published sources to support reproducibility.

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